Ghaf crown mapping

Quick reference

Everything here is expanded in the technical manual; the numbers in the right-hand column point at its chapters. Values come from `docs/handover/FACTS.yml`, which wins over this sheet if they ever disagree. Every command below assumes the `ghaf` environment is active, the working directory is the code folder, and `MODEL` names the FastViT model folder.

Prove it works — in this order

Command	Passes when	Ch.
`python -m pytest tests\ -q`	`336 passed` and no `F` — the code is intact. The number skipped varies with the machine	4
`python tools\smoke_test.py --checkpoints ..\models`	six rows of `ok`, every tensor matched	4
`python tools\check_dataset.py ..\data\ghaf --sample 0`	7005 / 869 / 767 paired	4

`0.00% ghaf` in the prediction column of the second command is correct: that forward pass runs on a blank synthetic tile, not on imagery.

Map one orthomosaic

```
python -m ghaf.inference.large_image ^
%MODEL%\fastvit-ma36_mask2former.py ^
%MODEL%\best_mIoU_iter_3500.pth ^
..\samples\Kalba26_sample.tif ^
--out-mask ..\output\crowns.tif ^
--out-polygons ..\output\crowns.gpkg ^
--batch-size 4 --min-area 1
```

> **THE DEFAULTS ARE NOT THE SETTINGS THAT PRODUCED THE RESULTS**
>
> Batch size defaults to 1, and `--min-area` to 0.0 — which keeps every single-pixel fragment as a crown and inflated the count on the sample clip from 16 to 27. `--out-polygons` will not run without `--out-mask` or `--out-prob`, and `--tile` **must stay 1024**: change it and the run still succeeds, still writes all three outputs, and reports a canopy figure produced at the wrong scale. *(Manual §6.5)*

Map a folder of images

```
python tools\predict_folder.py ^
%MODEL%\fastvit-ma36_mask2former.py ^
%MODEL%\best_mIoU_iter_3500.pth ^
D:\my-images --out-dir ..\output\folder ^
--batch-size 4 --min-area 1 --limit 3
```

Drop `--limit 3` once three images have come out right. One unreadable file does not stop the run: it is recorded in `summary.json` and the exit status is non-zero, neither of which is visible to somebody watching the progress bar. *(Manual §7.2)*

The numbers

	Value		Value
Tile	1024 px	Trained GSD	2.68 cm/px
Overlap	512 px	Tile on the ground	27.44 m (0.075 ha)
Threshold	0.5	Scratch needed	9 bytes per source pixel
Batch used here	4	Sample clip	8192 × 8192, 225 windows
Canopy, clip	0.9687 %	Canopy, test split	3.44 %
Crowns, clip	27 raw, 16 filtered	Crown area	2.4 to 112 m²

Deployed model mIoU **79.32**, F1 **87.22** on `testing/ghaf26`. Identify a checkpoint by its digest, not its filename: `f26cd5257b55058f…` for `best_mIoU_iter_3500.pth`, 252,650,755 bytes. The full digests are in `ghaf/release.py`.

Error → cause

What you see	Actually means	Ch.
`No module named mmengine`	wrong Python; `conda activate ghaf`	13
`mmseg ... is not installed in conda environment`	wrong environment; the message names the interpreter	13
`No module named ftfy`	`pip install ftfy regex`	13
`Numpy is not available`	NumPy 2 beside torch 1.12; pin `<2`	13
`CUDA out of memory`	lower `--batch-size`; **never** `--overlap`	13
`not enough scratch space`	`--scratch-dir` at a bigger disk	13
`--out-polygons needs --out-mask or --out-prob`	polygons are traced from a written raster	13
`path specified` printed twice	an `&` in the path; quote it	13
`no such folder ... D:\ghaf-project`	the manual's example path; use your own	13
`SHA-256 mismatch`	that checkpoint is not the released file	13
`0.00% ghaf` from `smoke_test.py`	correct; a blank tile	4
`0.00%` canopy on real imagery	wrong checkpoint, or bands not RGB	13
a crown count far too high	fragments; use `--min-area 1`	13
high mIoU, predicts nothing	masks encoded 0 and 255, not 0 and 1	13

Rules that are not negotiable

Rule	Because
Never edit `configs/_base_/ghaf.py`	all six models inherit it; a change silently redefines what every published score means
Keep `work_dirs` logs	no seed is set, so the seed of a run exists only in its log
`--load-from` to fine-tune, never `--resume`	`--resume` continues a schedule; `--load-from` starts a new one
Score a fine-tuned model on the old split too	it can gain on the new site and lose on the original, and nothing reports it
Keep companion files beside their tiles	moving a PNG without its `.pgw` strips the position silently
`check_dataset.py` before any training run	masks encoded 0 and 255 train to a high mIoU and predict nothing

Never established, so do not quote it: how long a full-mosaic run takes — the 84,072 × 103,691 mosaic has never been run end to end, and its 78.5 GB of scratch and 33,128 windows are arithmetic. Also unmeasured: training time, validation scores for the five models other than FastViT-MA36, and how often touching crowns merge into one polygon. *(Manual §14.3)*